

TIAV Multimodal App — Production Deployment Guide

***Scope:** Unified guide for deploying the TIAV FastAPI backend to Railway/Render and the React/Vite frontend to Vercel. **Last updated:** May 2026*

Architectural Diagram

```
graph TD
    A["[ React Frontend ] (Deployed on Vercel)"] -- "(HTTP POST / JSON / Stream)" --> B["[ FastAPI Backend Server ] (Deployed on Railway / Render)"]
    B -- "(External API Calls)" --> C["[ OpenRouter ]"]
    C -- "(LLM Inference)" --> D["[ Switchable Models ] (Qwen / Claude / GPT / etc.)"]
```

Part 1: Frontend Deployment (Vercel)

The React frontend compiles down to static assets using **Vite**.

1. Environment Variable to Set in Vercel

To point your production frontend to your deployed backend instead of `localhost`, you must configure a specific environment variable in the Vercel dashboard:

- **Variable Name:** `VITE_API_BASE_URL`
- **Variable Value:** `https://your-backend-url.railway.app` (replace with your active backend endpoint URL)

*[!IMPORTANT] - Vite automatically embeds variables starting with `VITE_` into the production client bundle during the build step. - **Do not add a trailing slash** at the end of the URL (e.g. use `https://tiav-backend.up.railway.app` , not `https://tiav-backend.up.railway.app/`).*

2. Deployment Steps

1. Push your repository to GitHub.
2. Log into the [Vercel Dashboard](#).
3. Click **Add New** → **Project**, and select your Git repository.
4. Set the following project configurations:
5. **Framework Preset:** Vite
6. **Root Directory:** `frontend`
7. Expand the **Environment Variables** section and input:
8. Key: `VITE_API_BASE_URL`
9. Value: `https://your-backend-url.railway.app`
10. Click **Deploy**. Vercel will build the frontend and serve it at a public `https://*.vercel.app` domain.

● Part 2: Backend Deployment (Railway or Render)

The backend is built in **FastAPI** and uses **uvicorn** for high-throughput ASGI serving.

1. Deployment Requirements

Make sure the `requirements.txt` file is present in the `backend/` directory so the hosting provider can install all dependencies. Your `backend/requirements.txt` currently contains:

```
fastapi
uvicorn
requests
python-multipart
```

Additionally, ensure a dynamic port binding command is used since Railway and Render inject a random `$PORT` environment variable.

2. Environment Variables to Set

To secure CORS and connect to NVIDIA's NIM endpoints:

1. **ENV** : Set to `production` . (Configures backend CORS to allow all origins `["*"]` dynamically).
2. **ALLOWED_ORIGINS** (Optional): A comma-separated list of exact allowed domain origins (e.g., `https://your-frontend.vercel.app`) if you wish to lock down cross-origin requests instead of using `"*"` .
3. **PORT** : Will be injected automatically by Render or Railway.

3. Deployment Steps (Railway — Recommended)

1. Go to the [Railway Dashboard](#).
2. Click **New Project** → **Deploy from GitHub repo**.
3. Select your repository.
4. Set the **Root Directory** or **Build Command** configurations:
5. **Root Directory:** `backend`
6. **Start Command:** `uvicorn main:app --host 0.0.0.0 --port $PORT`
7. Under **Variables**, add:
8. **ENV** : `production`
9. Railway will automatically build, provision an SSL certificate, and assign a public domain (e.g., `https://backend-production.up.railway.app`).

4. Deployment Steps (Render)

1. Go to the [Render Dashboard](#).
2. Click **New** → **Web Service** and connect your GitHub repo.
3. Set the following details:
4. **Root Directory:** `backend`
5. **Runtime:** Python
6. **Build Command:** `pip install -r requirements.txt`
7. **Start Command:** `uvicorn main:app --host 0.0.0.0 --port $PORT`
8. Under **Advanced** → **Environment Variables**, add:
9. **ENV** : `production`
10. Click **Create Web Service**.

Production Testing & Verification

Once both frontend and backend are successfully deployed:

1. Go to your frontend Vercel URL (e.g., `https://tiav-multimodal-studio.vercel.app`).
2. Type a message in the chat box and press **Enter**.
3. The network tab will make cross-origin requests to `https://your-backend-url.railway.app/text` .
4. If there is a connection issue, the chat box will immediately display:

"Failed to connect to the backend server. If in production, ensure your VITE_API_BASE_URL environment variable is set to your active backend address."

5. Once working, standard tokens will stream smoothly word-by-word into conversational ChatGPT bubbles!